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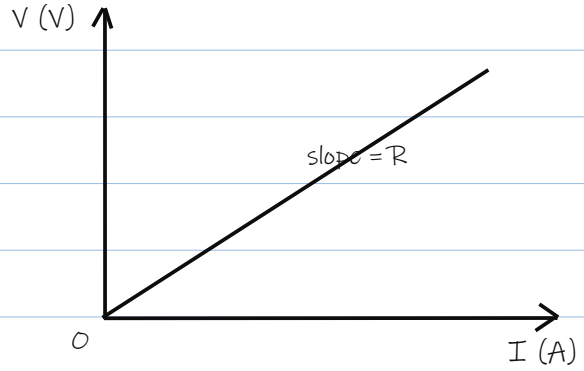
Subject: Engineering Physics - IA 1 | Section: A

Ans 3 (answering this first)

Ohm's law: The current through a conductor is directly proportional to the voltage across it, provided the temperature stays constant. $V = IR$.



Resistance = slope of the V-I line = V/I.

Ans 1

~~First law: F = ma~~

First law: a body stays at rest or in uniform motion unless an external force acts on it.

Second law: F = ma, rate of change of momentum is proportional to force.

Third law: action and reaction are equal and opposite, e.g. when we walk we push the ground back and the ground pushes us forward.

Ans 2

~~a = 20 x 5 = 100~~

a = (v - u)/t = 20/5 = 4 m/s^2

for distance, s = ut + 1/2 at^2 = 1/2 x 4 x 25

(Q4 not attempted - ran out of time)